

JKBOSE 2026

CLASS 11TH

Bhaichara
ON
TOP.

#XUV 

ORGANIC CHEMISTRY:

SOME BASIC PRINCIPLES AND TECHNIQUES

LECTURE-04

CHEMISTRY

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Topics To Covered

1

IUPAC Nomenclature

REVISION



2° Prefix + 1° Prefix + Word Root + 1° suffix + 2° suffix

Substituent + Cyclo + No. of
C-atoms
in PCC. + ane
or
ene
or
yne + FG.

1) Decide the PCC

= Max. FG > Max no. of (= / $\equiv$) > Max. no. of C-atoms > Max no. of Substituent > Alphabetical order of Substituent

2) Numbering of PCC.

= Lowest no. FG > Lowest no. (= / $\equiv$) > Lowest no. Substituent > Alphabetical order of Substituent.

3) Naming

IUPAC Nomenclature

1. Priority Order For Deciding Parent Chain

- (a) Maximum number of functional groups.
- (b) Maximum number of multiple bonds.
- (c) Maximum number of carbon atoms in main chain.
- (d) Maximum number of substituents.
- (e) Lowest set of locant (FG > *c-c* multiple bonds > substituents).
- (f) Alphabetical order of name of substitute.

IUPAC Nomenclature

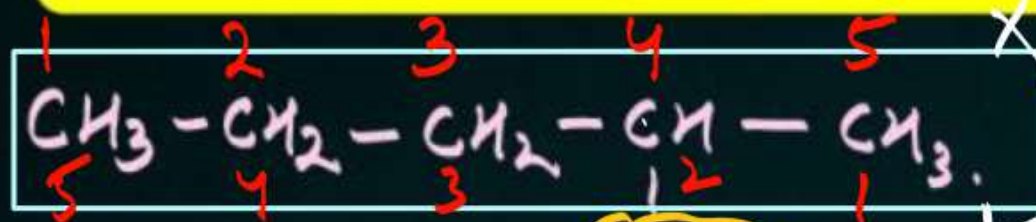
2. Priority Order For Numbering of Parent Chain

- (a) Lowest possible number is given to the functional group.
- (b) Lowest possible number is given to carbon multiple Bond.
- (c) Lowest possible number is given to the substituent
- (d) Lowest possible number is given according to alphabetical order of substituents.

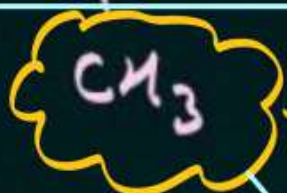
Subst + cycle + wk + ^{ane}ene + FG.

[illegible]

IUPAC Nomenclature



PCC



Substituent

2-Methylpentane

F = Fluoro

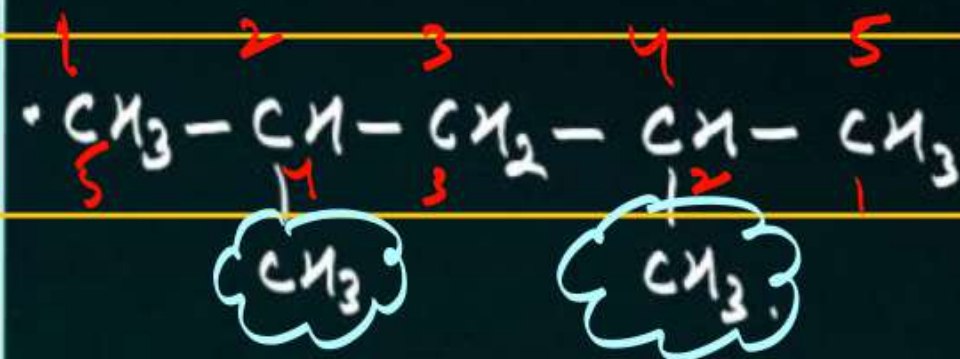
Cl = Chloro

I = Iodo

$-\text{O}-\text{CH}_3$ = methoxy

NO_2 = Nitro

Sub + Cyclo + WR + one or en or ine + F.G.

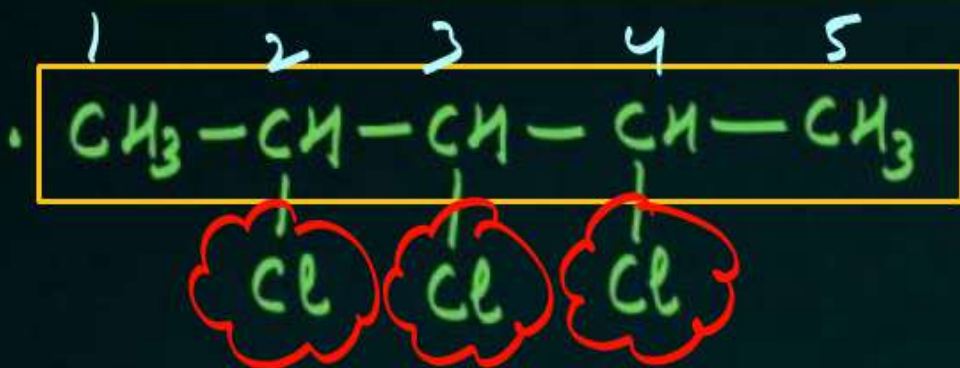


2,4-Dimethylpentane

Rule: → If there are more than one same substituent in the PCC, then we use.
 di for two
 tri for three
 tetra for four
 penta for five and so on.



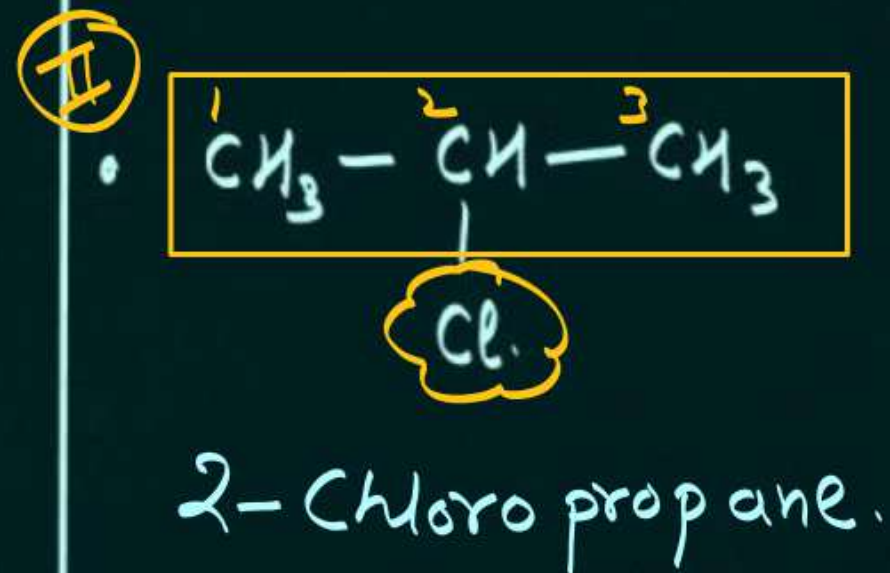
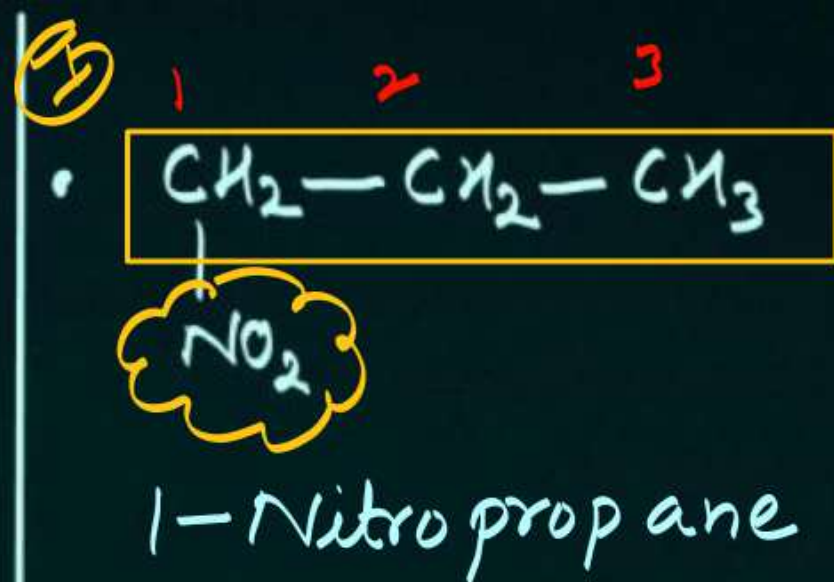
IUPAC Nomenclature



2,3,4-Trichloropentane

Tawkeer
Amin
Safi

Substituent + cyclo + WR + ane or ene or yne + FG.



1) Decide the PCC

= $\begin{matrix} \text{Max.} \\ \text{FG} \end{matrix} \begin{matrix} \text{Max no. of} \\ (= / \equiv) \end{matrix} \begin{matrix} \text{Max. no. of} \\ \text{C-atoms} \end{matrix} \begin{matrix} \text{Max no. of} \\ \text{Substituent} \end{matrix} \begin{matrix} \text{Alphabetical} \\ \text{order} \\ \text{of Substituent} \end{matrix}$

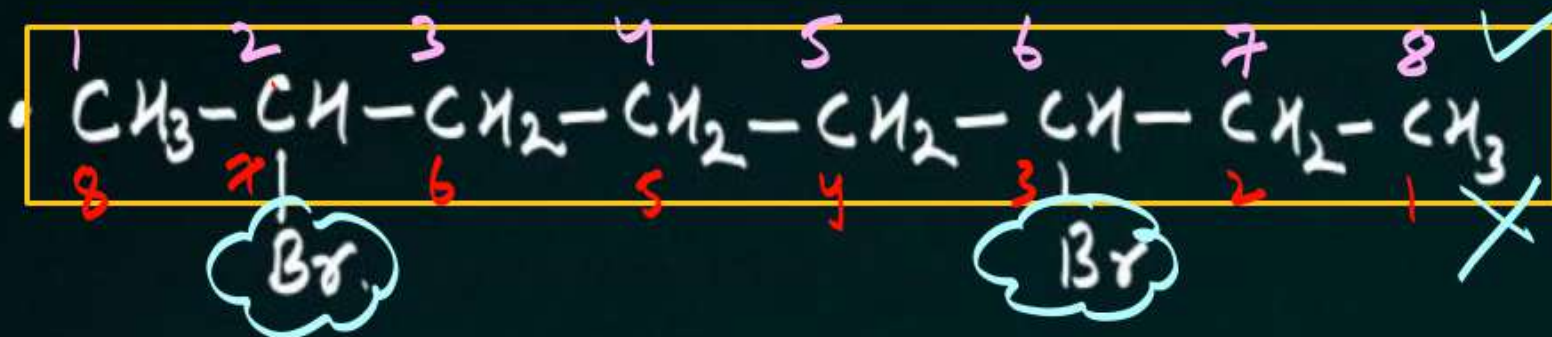
2) Numbering of PCC.

= $\begin{matrix} \text{Lowest no.} \\ \text{FG} \end{matrix} \begin{matrix} \text{Lowest no.} \\ (= / \equiv) \end{matrix} \begin{matrix} \text{Lowest no.} \\ \text{Substituent} \end{matrix} \begin{matrix} \text{Alphabetical} \\ \text{order of} \\ \text{Substituent.} \end{matrix}$

3) Naming

Substituent + Cyclo + WR + $\begin{matrix} \text{ane} \\ \text{or} \\ \text{ene or yne} \end{matrix}$ + FG.

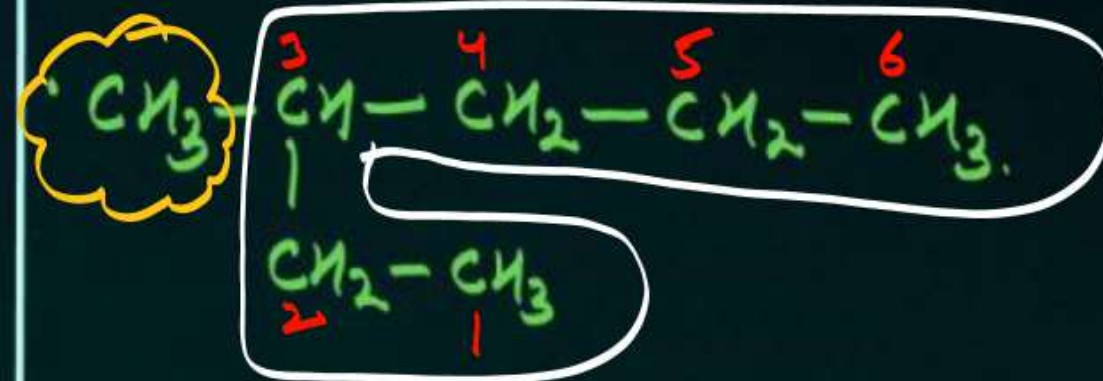
IUPAC Nomenclature



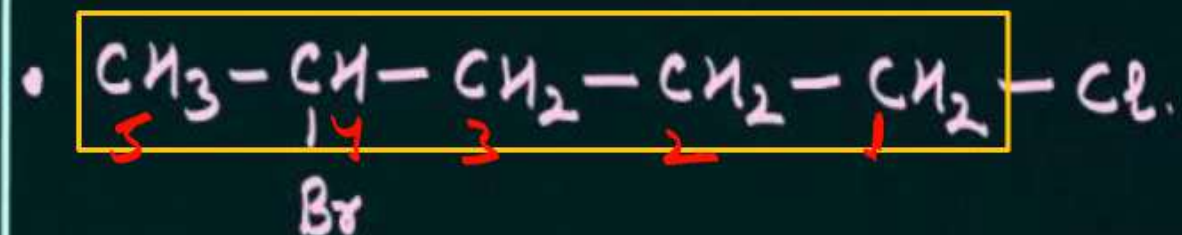
2, 6 - Dibromooctane

FG > MB > Substituent

Substituent + Cyclo + WR +
 and
 or ene
 or yne + FG



3-Methylhexane

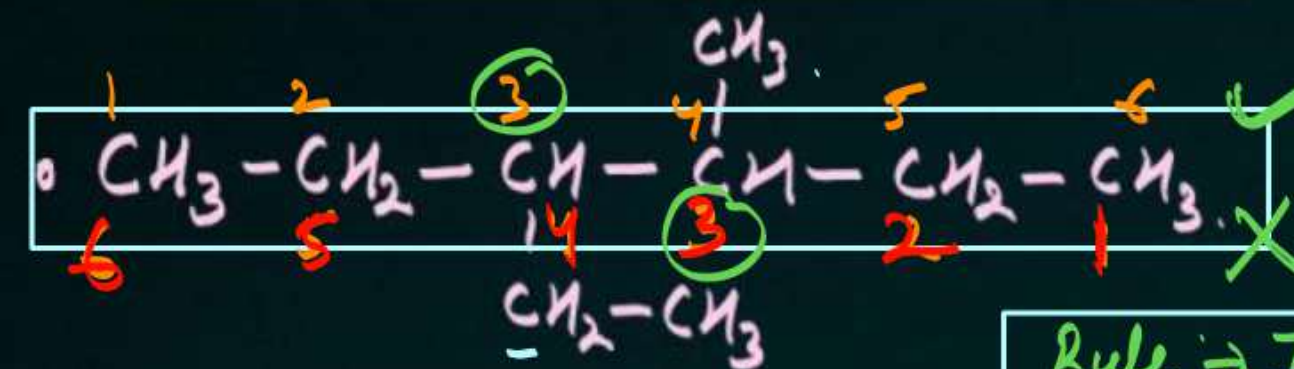


Two name possible

- 1-Chloro-4-bromopentane.
- 4-Bromo-1-chloropentane. ✓

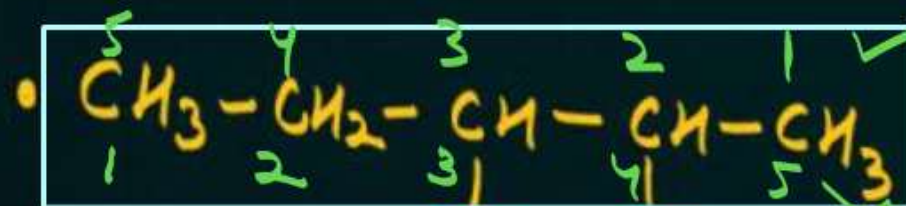
Rule: if there are more than one different substituents then naming is done alphabetically according to the name of substituent.

IUPAC Nomenclature

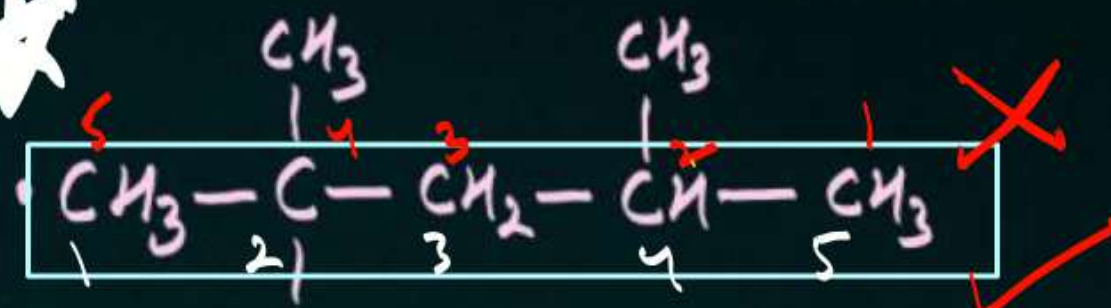


3-Ethyl-4-Methylhexane

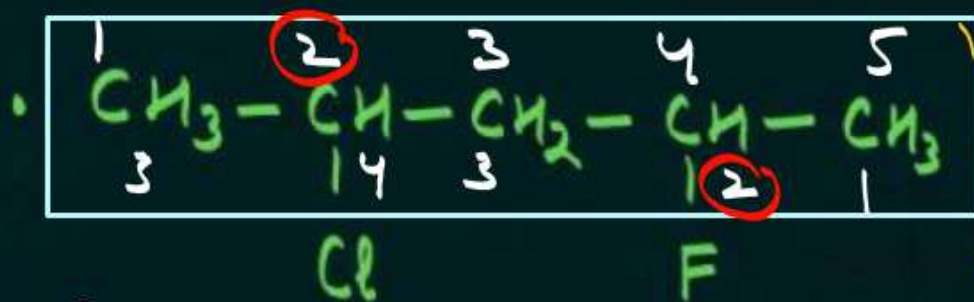
Rule: $\rightarrow$ If in different substituents there is tie in numbering, then numbering is done alphabetically.



2-Chloro-3-fluoropentane

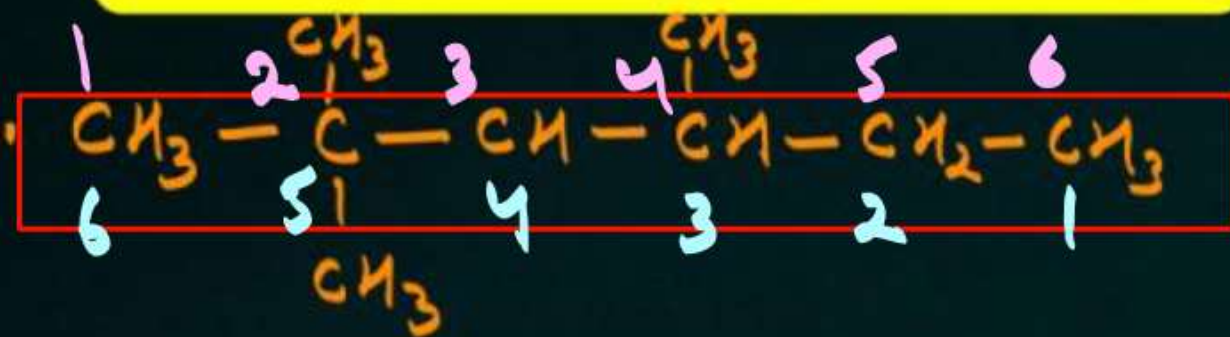


$(2, 4, 4)$
 $(2, 2, 4)$ $\rightarrow$ Lowest set of locant
 2, 2, 4-Trimethylpentane



2-Chloro-4-fluoropentane

IUPAC Nomenclature



2, 2, 4 ✓
X

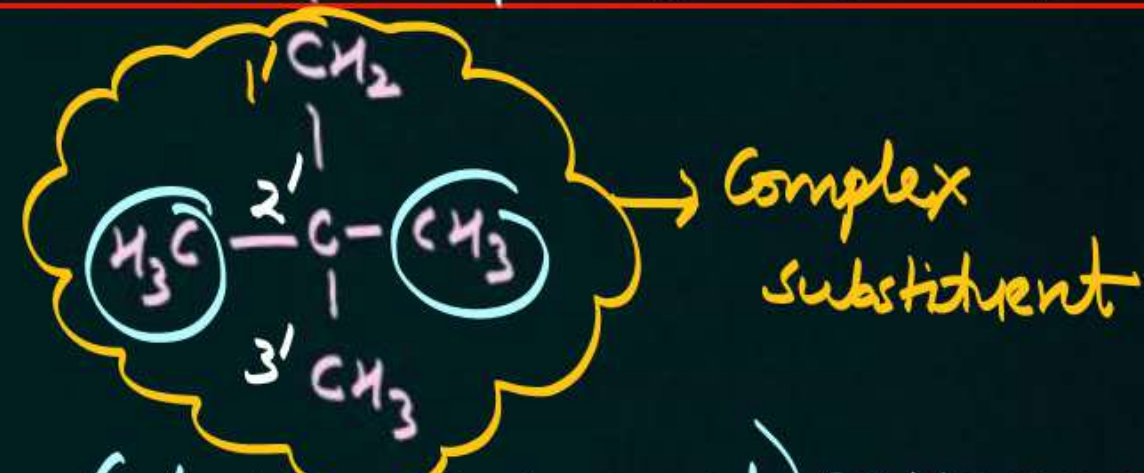
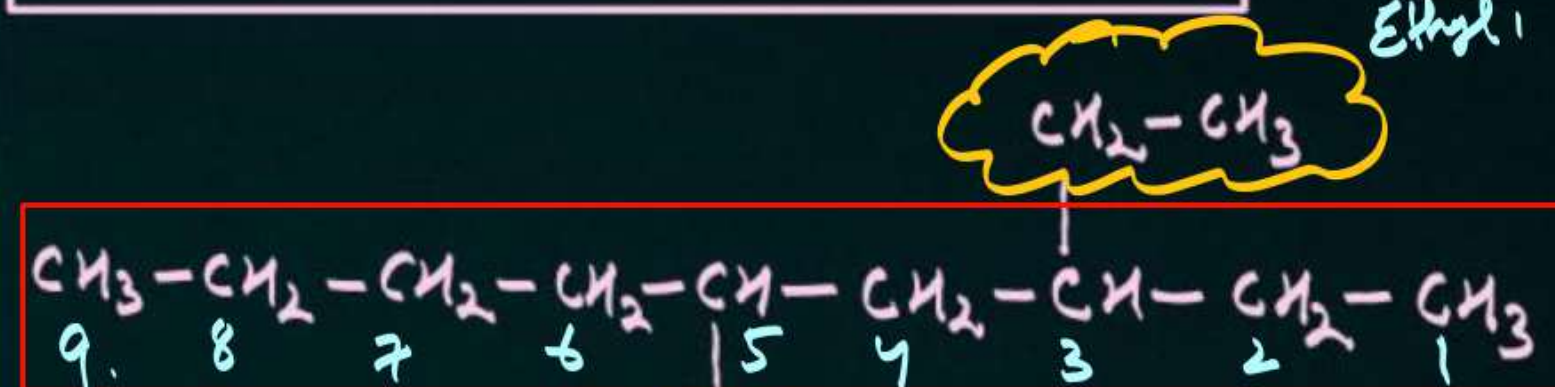
2, 2, 4-Trimethylhexane.

Subst + Cyclo + WR + ane
or ene
or yne + FG.

NAMING OF COMPLEX SUBSTITUENT.



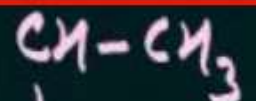
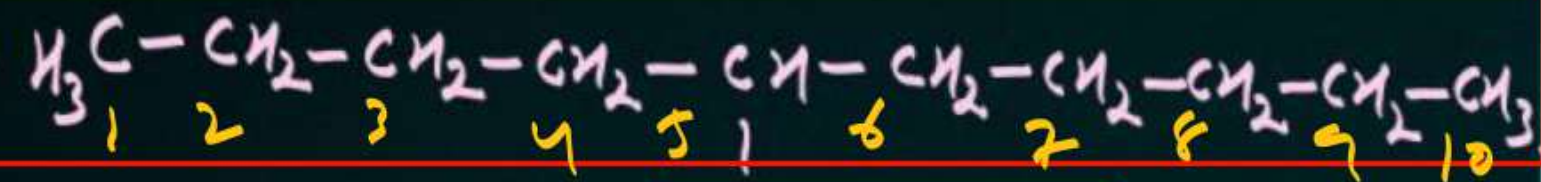
Ethyl



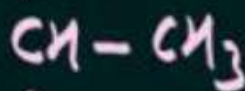
3-Ethyl-5-(2,2-dimethylpropyl)nonane.

In case of complex substituent $\Rightarrow$ C-attached with PCC is numbered as 1' and then different IUPAC is applied on that & name of substituent is ended with 'yl'.
also written in bracket.

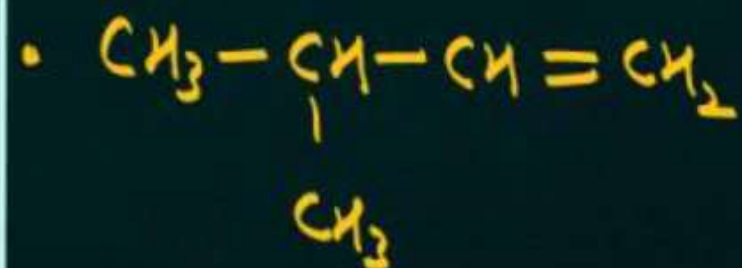
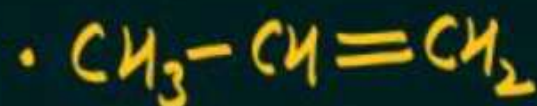
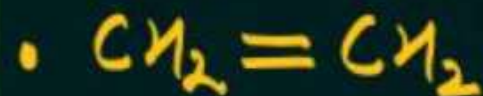
IUPAC Nomenclature



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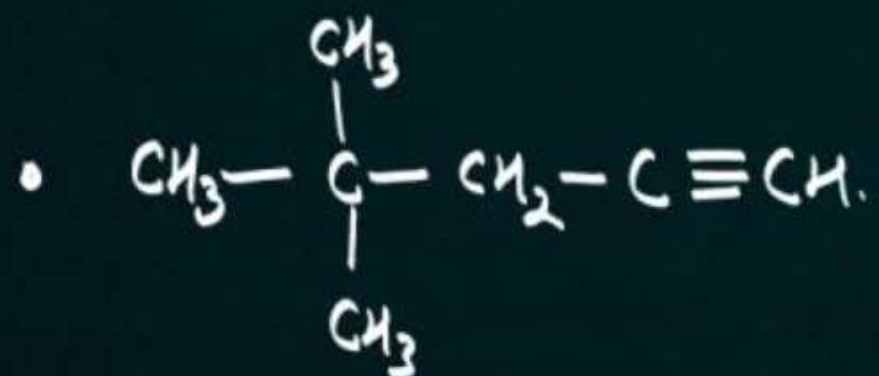
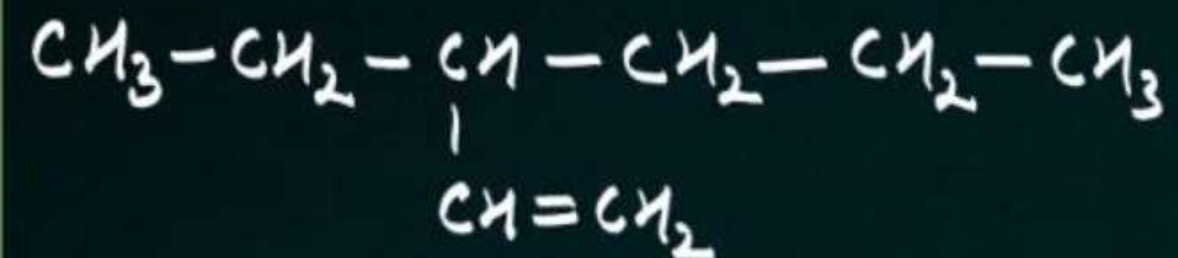


2. Nomenclature of Alkenes or Alkynes

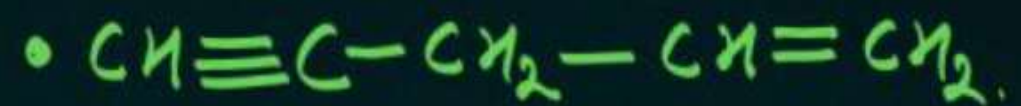


Home
Work

IUPAC Nomenclature



IUPAC Nomenclature



QUESTION

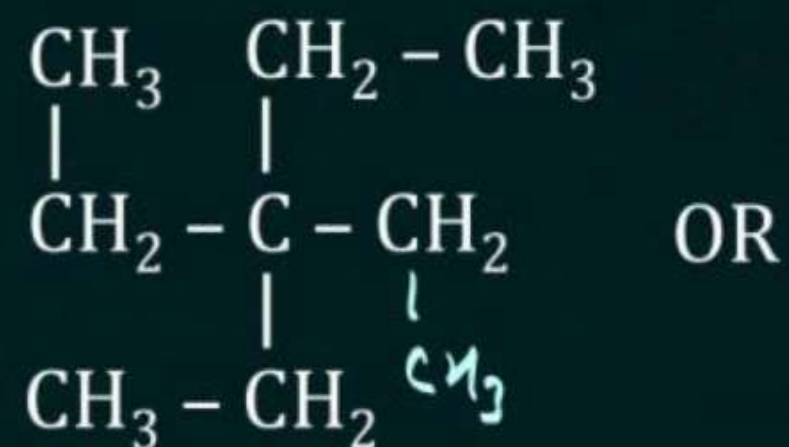
Correct IUPAC name of following compound

1 1, 2, 3-dimethyl pentane.

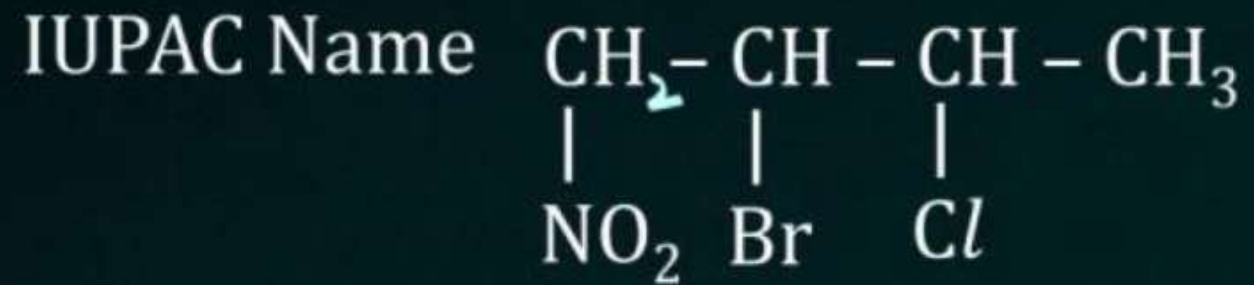
2 3, 2-diethyl pentane

3 3, 3 - diethyl pentane

4 2, 3-diethyl pentane



QUESTION



- 1 2-Chloro-3-Bromo-4-nitrobutane
- 2 2-Bromo-3-Chloro-1-nitrobutane
- 3 1-Nitro-2-Bromo-3-chloro butane
- 4 4-Nitro-3-Bromo-2-chloro butane

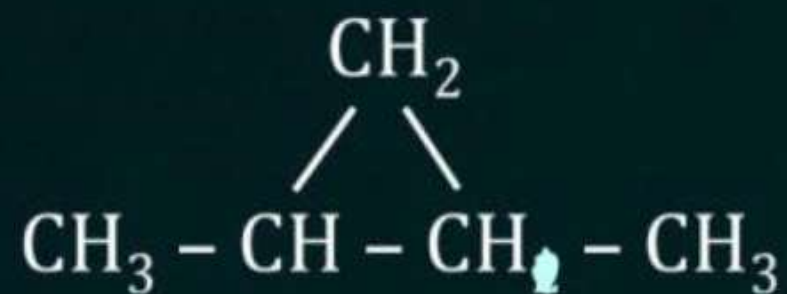
QUESTION



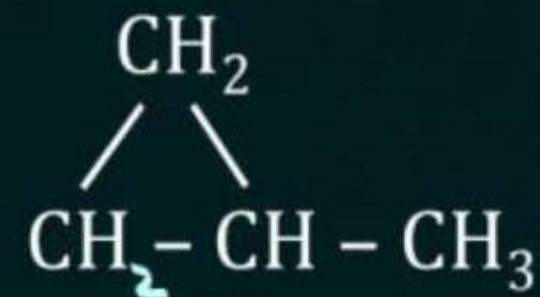
Correct Representation of following compound is



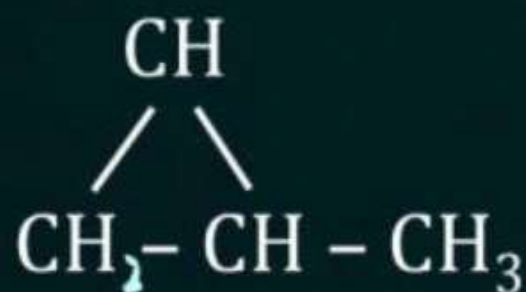
1



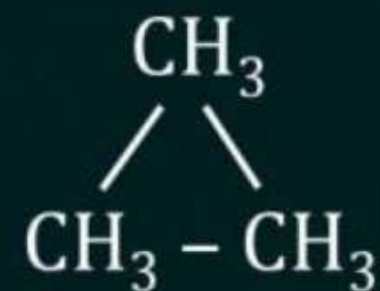
2



3

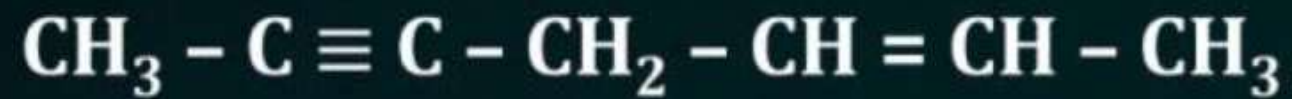


2



QUESTION

IUPAC name of following compound is



- 1 Oct-2-yne-5-ene
- 2 Hept-2-yne-5-ene
- 3 Hept-5-yne-2-ene
- 4 Hepta-5, 2-diene

Thank
You